

Bench Protocols: M1/M2 Macrophage Contributions to Colon Cancer DTC Dormancy

Step-by-Step Instructions for Postdoctoral Researchers

May 2026

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Document Overview

This document provides **complete, step-by-step bench protocols** for all experimental procedures in the study: “Quantitative Dissection of M1 and M2 Macrophage Contributions to Dormancy Entry, Maintenance, and Exit in Colon Cancer.”

Each protocol includes: materials list, detailed steps, timing, expected results, and troubleshooting tips.

TIP

Protocols are organized in experimental execution order. Complete Protocol 1 (cell engineering) before starting Protocol 2 (in vivo work).

Protocol 1: MC38 Cell Line Engineering

1.1 Purpose

Engineer MC38 murine colon adenocarcinoma cells with fluorescent reporters for dormancy detection and longitudinal tracking.

1.2 Required Constructs

Construct	Reporter	Purpose	Lentiviral Vector
GFP-Luciferase	EGFP + firefly luciferase	BLI tracking + flow cytometry	pLenti-PGK-GFP-2A-Luc2
Fucci2a	mCherry-hCdt1 (G1) + mVenus-hGem (S/G2/M)	Real-time cell cycle	pRRL-Fucci2a
H2B-mCherry	Histone H2B-mCherry	Nuclear label for single-cell tracking	pLenti-EF1a-H2B-mCherry

1.3 Lentivirus Production (Day 0-3)

Materials:

- HEK293T cells (ATCC CRL-3216)
- Lentiviral packaging plasmids: psPAX2, pMD2.G
- Transfer plasmids (listed above)
- Lipofectamine 3000 (ThermoFisher L3000015)
- DMEM + 10% FBS + 1% Pen/Strep
- 0.45 μ m PVDF filter units
- Lenti-X Concentrator (Takara 631232)

Steps:

1. **Day 0:** Seed HEK293T cells at 5×10^6 cells per 10 cm dish in 10 mL DMEM + 10% FBS. Incubate 37°C, 5% CO₂ overnight.
2. **Day 1 (morning):** Confirm ~80% confluence. Prepare transfection mix per dish:
 - Tube A: 375 μ L Opti-MEM + 10 μ g transfer plasmid + 7.5 μ g psPAX2 + 2.5 μ g pMD2.G + 50 μ L P3000 reagent
 - Tube B: 375 μ L Opti-MEM + 60 μ L Lipofectamine 3000
 - Combine Tube A + Tube B. Incubate 15 min at RT. Add dropwise to dish.
3. **Day 2:** Replace medium with 10 mL fresh DMEM + 10% FBS + 1% BSA (improves viral stability).
4. **Day 3:** Collect supernatant. Filter through 0.45 μ m PVDF. Add Lenti-X Concentrator (1:3 ratio). Incubate 4°C overnight. Centrifuge 1500 x g, 45 min, 4°C. Resuspend pellet in 200 μ L PBS. Aliquot and store at -80°C.

WARNING

Biosafety Level 2 (BSL-2) required for lentiviral work. Use biosafety cabinet for all steps involving virus.

1.4 MC38 Transduction (Day 0-14)

Steps:

1. Seed MC38 cells at 2×10^5 per well in 6-well plate (DMEM + 10% FBS).
2. Add concentrated lentivirus (GFP-Luc2 construct first) at MOI ~5 with 8 ug/mL polybrene.
3. Spin-infect: Centrifuge plate at 800 x g, 30 min, 32°C.
4. Replace medium after 12 h. Culture for 48 h.
5. Sort GFP+ cells by FACS (BD FACSAria or equivalent). Gate: FSC/SSC singlets -> Live (DAPI-) -> GFP+. Collect top 30% brightest GFP+ cells.
6. Expand sorted cells for 5-7 days.
7. Repeat transduction with Fucci2a construct (same MOI). Sort dual-positive cells: GFP+ AND mCherry+ (Fucci2a).
8. Repeat with H2B-mCherry construct.
9. **Final validation:** Culture triple-reporter cells for 2 weeks. Confirm stable GFP, Fucci2a signal (cell cycle-dependent mCherry/mVenus), and nuclear mCherry (H2B) by fluorescence microscopy. Confirm luciferase activity with D-luciferin assay.

TIP

Freeze multiple vials of validated triple-reporter MC38 cells at each passage. Create a master cell bank at Passage 3-5 post-engineering.

Expected result: >90% GFP+, functional Fucci2a cycling (red in serum-starved quiescence, green in log-phase proliferation), nuclear mCherry in >95% cells.

Protocol 2: Intrasplenic Injection — Liver Metastasis Model

2.1 Purpose

Introduce MC38-GFP-Luc-Fucci2a-H2B-mCherry cells into the liver via the splenic-portal vein route to generate disseminated tumor cells (DTCs).

2.2 Materials

- C57BL/6J mice, female, 8-10 weeks (Jackson Labs #000664)
- MC38-GFP-Luc-Fucci2a cells (from Protocol 1)
- Isoflurane vaporizer and induction chamber
- Surgical microscope or loupe magnification
- 30G insulin syringes (BD Ultra-Fine)
- Microsurgical instruments: fine forceps (Dumont #5), micro-scissors, needle holder
- 5-0 Vicryl sutures (absorbable, for peritoneum)
- Wound clips (9 mm Autoclip) + clip applier
- Buprenorphine (0.3 mg/mL) for analgesia
- Sterile saline, sterile gauze, sterile drapes
- Heat pad (maintain 37°C during surgery)

2.3 Cell Preparation (Day of Surgery)

1. Harvest MC38-GFP-Luc-Fucci2a cells at 70-80% confluence (use cells at passage <20 post-engineering).
2. Wash with PBS. Trypsinize with 0.05% trypsin-EDTA, 3 min at 37°C.
3. Neutralize with DMEM + 10% FBS. Centrifuge 300 x g, 5 min.
4. Wash 2x with ice-cold PBS. Resuspend at **1 × 10³ cells per 50 uL PBS** (low dose for dormancy) or 1 × 10⁵ cells per 50 uL (high dose for positive control).
5. Keep on ice. Use within 30 minutes of preparation.

WARNING

Cell viability must be >95

2.4 Surgical Procedure

1. **Anesthesia:** Place mouse in induction chamber. Isoflurane 3% in 1 L/min O₂. After induction (~2 min), transfer to nose cone on heated surgical platform. Maintain at 1.5-2% isoflurane.
2. **Preparation:** Apply ophthalmic ointment to eyes. Shave left flank. Disinfect with betadine followed by 70% ethanol, x3 alternating.
3. **Incision:** Make 1 cm left flank incision through skin and peritoneum.
4. **Spleen exposure:** Gently exteriorize spleen using moistened cotton swab. Place on sterile gauze moistened with warm saline.
5. **Injection:** Using 30G insulin syringe, slowly inject **50 uL cell suspension** (1 × 10³ cells) into the inferior pole of the spleen over **30 seconds**. Too-fast injection causes backflow and peritoneal seeding.
6. **Wait:** Hold needle in place for 60 seconds after injection to prevent back-bleeding.
7. **Withdraw** needle gently. Apply gentle pressure with cotton swab to injection site.
8. **Splenectomy (optional but recommended):** Ligate splenic vessels with 5-0 silk suture. Remove spleen to prevent local tumor growth at injection site. This step is critical for eliminating confounding primary splenic tumors.
9. **Closure:** Close peritoneum with 5-0 Vicryl (continuous suture). Close skin with wound clips (2-3 clips).
10. **Recovery:** Place mouse in clean cage on heat pad. Monitor until ambulatory (typically 15-30 min).

11. **Analgesia:** Buprenorphine 0.1 mg/kg SC every 12 h for 48 h post-surgery.

Post-operative monitoring: Daily health checks for 7 days. Remove wound clips at Day 10-14.

TIP

Practice the injection technique on cadaver mice first. The key is SLOW injection (30 sec) to allow cells to transit the splenic vein into the portal circulation before back-pressure builds up.

2.5 Expected Timeline Post-Injection

Day	Phase	Expected Biology
0-1	Seeding	DTCs transit portal vein, lodge in liver sinusoids
1-3	Early extravasation	DTCs exit sinusoids into liver parenchyma
3-7	Dormancy entry	DTCs either enter dormancy or begin proliferating
7-14	Dormancy establishment	Stable NR2F1+ Ki67- DTCs present in liver
14-84	Maintenance / Exit	Variable: some DTCs remain dormant, others exit

Protocol 3: Macrophage Depletion and Polarization Manipulation

3.1 Total Macrophage Depletion — Clodronate Liposomes

Reagent: Clodronate liposomes (Liposoma BV, Amsterdam, Cat# CP-005-005)

Dosing schedule:

Timing	Dose	Route	Frequency
3 days before injection	200 uL	IV tail vein	Single
Day 1 onward	200 uL	IV tail vein	Twice weekly (e.g., Mon + Thu)

Control: PBS liposomes (Liposoma BV, Cat# PB-005-005), same volume and schedule.

Preparation:

1. Remove clodronate liposomes from 4°C storage. **Do NOT freeze.**
2. Gently invert tube 5x to resuspend. **Do NOT vortex** (damages liposomes).
3. Draw 200 uL into 1 mL syringe with 30G needle.
4. Warm to room temperature before injection.

Injection technique (tail vein):

1. Place mouse in restraint tube. Warm tail with heat lamp or warm water soak for 1-2 min to dilate veins.
2. Identify lateral tail vein. Clean with 70% ethanol.
3. Insert 30G needle at 10-20° angle, bevel up, 2-3 cm from tail base.
4. Inject slowly over 15-20 seconds. No resistance should be felt.
5. Withdraw needle, apply gentle pressure with gauze.

WARNING

Depletion efficiency must be verified at each sacrifice timepoint by flow cytometry. Target: >80

3.2 Kupffer Cell-Specific Depletion — CD169-DTR Mice

Mouse strain: CD169-DTR (B6.FVB-Tg(Siglec1-DTR/EGFP)1Mnz/J)

Reagent: Diphtheria toxin (DT), Sigma D0564

DT preparation:

1. Reconstitute DT in sterile PBS to 1 mg/mL stock. Aliquot 10 uL. Store at -80°C.
2. Working solution: Dilute stock to 1 ug/mL in sterile PBS on day of use.
3. **Dose:** 4 ng/g body weight IP. Example: 25 g mouse = 100 uL of 1 ug/mL solution.

Schedule:

- **Day -2:** First DT injection (48 h before tumor cell injection)
- **Day 0:** Tumor cell injection
- **Day 2, 5, 8, 11, ...:** Repeat DT every 72 h for maintenance

Depletion verification: At sacrifice, stain liver non-parenchymal cells for TIM4+ CD11c+ (Kupffer cells). Target: >80% depletion of TIM4+ cells.

WARNING

DT is EXTREMELY toxic. The therapeutic window is narrow. Do NOT exceed 10 ng/g. Include wild-type + DT control mice to control for DT off-target effects.

3.3 M1 Polarization — CpG-ODN 1826

Reagent: CpG-ODN 1826 (InvivoGen tlr1-1826), mouse-specific TLR9 agonist

Preparation: Resuspend in sterile endotoxin-free H₂O to 1 mg/mL. Aliquot and store at -20°C.

Dose: 50 ug per mouse IP, 3 times per week (e.g., Mon/Wed/Fri)

Schedule: Begin Day -3 (3 days before injection). Continue throughout experiment.

Expected effect: Shifts liver macrophages toward M1-like phenotype. Verify by flow cytometry: increased CD86+ MHC-II-high fraction among F4/80+ cells at Day 7.

3.4 M2 Polarization — IL-4 Immune Complex**Reagents:**

- Recombinant mouse IL-4 (BioLegend 574302)
- Anti-IL-4 antibody, clone 11B11 (BioLegend 504111)

Complex preparation (prepare fresh each time):

1. In 200 uL sterile PBS, mix 5 ug IL-4 + 25 ug anti-IL-4 antibody.
2. Incubate at 37°C for 30 min to form immune complexes.
3. Inject 200 uL IP immediately.

Schedule: Twice weekly (e.g., Tue + Fri). Begin Day -3 or Day 1 as per experimental arm.

Expected effect: Sustained M2 polarization. Verify by flow cytometry: increased CD206+ fraction among F4/80+ cells.

3.5 Anti-IFN-gamma Neutralization

Reagent: Anti-mouse IFN-gamma antibody, clone XMG1.2 (Bio X Cell BP0055)

Dose: 200 ug per mouse IP, twice weekly

Control: Rat IgG1 isotype control (Bio X Cell BP0088), same dose

3.6 Anti-IL-4Ralpha Blockade

Reagent: Anti-mouse IL-4Ralpha antibody, clone M1 (Bio X Cell)

Dose: 200 ug per mouse IP, twice weekly, starting Day 14

Purpose: Blocks both IL-4 and IL-13 signaling, preventing M2 skewing.

Protocol 4: Bioluminescence Imaging (BLI)

4.1 Purpose

Weekly monitoring of tumor burden to detect dormancy exit (BLI signal increase indicates proliferative outgrowth).

4.2 Materials

- IVIS Spectrum or Lumina III (PerkinElmer)
- D-Luciferin potassium salt (GoldBio LUCK-1G)
- Isoflurane anesthesia system compatible with IVIS

4.3 Procedure

1. **Prepare luciferin:** 15 mg/mL D-luciferin in sterile PBS. Filter-sterilize (0.22 μ m). Store aliquots at -20°C (stable for 6 months).
2. **Inject luciferin:** 150 mg/kg IP (e.g., 25 g mouse = 250 μ L of 15 mg/mL). Record exact time of injection.
3. **Wait 10 minutes** for peak luminescence. Place mice in IVIS under isoflurane anesthesia (1.5%).
4. **Image settings:**
 - Exposure: Auto (or start with 1 min, adjust if saturated)
 - Binning: Medium (8)
 - F/stop: 1
 - Emission filter: Open
5. **Acquire image.** Save as .nii or .tif. Record total flux (photons/sec) using Living Image software or open-source IVIS-compatible software.

4.4 Analysis

1. Draw region of interest (ROI) around the liver area for each mouse.
2. Record total flux (photons/sec) per ROI.
3. **Dormancy = stable or decreasing BLI signal for 4+ consecutive weeks.**
4. **Dormancy exit = >2-fold increase in total flux above baseline plateau.**
5. Export data to spreadsheet. Plot longitudinal BLI curves per mouse.

TIP

Dormant single DTCs are BELOW the BLI detection limit (approximately 10^3 – 10^4 cells). BLI can only detect proliferative outgrowth. A mouse with stable low BLI signal still harbors dormant DTCs that will be visible by IF on tissue sections.

Protocol 5: Liver Tissue Harvest and Processing

5.1 Purpose

Collect liver tissue at defined timepoints for immunofluorescence, flow cytometry, and molecular analysis.

5.2 Timepoints

Day	Phase	Mice per group	Analysis
7	Dormancy entry	3-4	IF, flow, cytokines
21	Dormancy established	3-4	IF, flow, scRNA-seq
42	Maintenance	3-4	IF, flow, scRNA-seq
84	Late/outgrowth	3-4	IF, flow, pathology

5.3 Euthanasia and Perfusion

- Euthanize by CO₂ asphyxiation followed by cervical dislocation.
- Portal vein perfusion (CRITICAL for flow cytometry):**
 - Make midline abdominal incision. Identify portal vein.
 - Insert 24G angiocatheter into portal vein. Secure with suture.
 - Cut inferior vena cava to allow outflow.
 - Slowly perfuse 10 mL ice-cold PBS over 2-3 minutes. Liver should blanch to pale tan color.
 - Purpose:** Removes circulating leukocytes that would contaminate liver-resident immune cell analysis.
- Excise liver. Record total weight and photograph.

5.4 Tissue Allocation

Divide each liver into portions:

Portion	Weight	Processing	Analysis
Left lobe (~30%)	~300 mg	Fix in 4% PFA, 24 h at 4°C	Immunofluorescence (Protocol 6)
Median lobe (~30%)	~300 mg	Fresh, on ice	Flow cytometry (Protocol 7)
Right lobe (~20%)	~200 mg	Snap-freeze in liquid N ₂	scRNA-seq (Protocol 8) or store -80°C
Caudate lobe (~20%)	~200 mg	Snap-freeze in liquid N ₂	Cytokine analysis, RNA, protein

WARNING

Process flow cytometry samples within 30 min of harvest. Cell viability drops rapidly after 1 hour on ice.

Protocol 6: Multiplex Immunofluorescence on Liver Sections

6.1 Purpose

Detect and quantify dormant DTCs, macrophage polarization, and macrophage-DTC spatial relationships in situ.

6.2 Tissue Sectioning

1. After 24 h PFA fixation, transfer liver tissue to 30% sucrose in PBS at 4°C until tissue sinks (24-48 h).
2. Embed in OCT compound. Freeze on dry ice. Store at -80°C.
3. Cut 8 um cryosections at -20°C. Mount on Superfrost Plus slides.
4. Air dry 30 min. Store at -80°C until staining.

6.3 Multiplex IF Panel — Dormancy + Macrophage

Channel	Marker	Purpose	Primary Antibody	Dilution
DAPI	Nuclear	Cell ID	-	-
Opal 520	pan-Cytokeratin	DTC identification	AE1/AE3 (Dako)	1:200
Opal 570	NR2F1	Dormancy marker	P1H2 (Santa Cruz)	1:100
Opal 620	Ki67	Proliferation	MIB-1 (Dako)	1:200
Opal 690	p27/KIP1	Cell cycle arrest	DCS-60.F6	1:200
Opal 780	CD68	Pan-macrophage	PG-M1 (Dako)	1:100

Alternative: serial sections for M1 vs M2 (on adjacent sections):

Section	Panel
Section A	GFP + F4/80 + Ki67 + DAPI (DTC detection + macrophage + proliferation)
Section B	GFP + CD86 + CD206 + DAPI (M1 vs M2 macrophage polarization)

6.4 Staining Procedure (Opal/tyramide method)

1. **Fix sections:** 4% PFA, 10 min, RT. Wash 3x PBS.
2. **Antigen retrieval:** Citrate buffer pH 6.0, microwave 20 min (or pressure cooker 125°C, 3 min). Cool 20 min. Wash 3x PBS.
3. **Block:** 5% normal goat serum + 0.3% Triton X-100 in PBS, 1 h, RT.
4. **Primary antibody incubation:** Dilute antibody in blocking buffer. Incubate overnight at 4°C in humidified chamber.
5. **Wash** 3x PBS-T (0.05% Tween-20), 5 min each.
6. **HRP-secondary:** Species-appropriate HRP-conjugated secondary, 1 h, RT.
7. **Wash** 3x PBS-T.
8. **Opal fluorophore:** Dilute Opal reagent 1:100 in amplification buffer. Incubate 10 min, RT.

9. **Wash** 3x PBS-T.
10. **Microwave strip:** Citrate buffer pH 6.0, microwave 20 min to remove antibodies while retaining deposited Opal fluorophore. Cool, wash.
11. **Repeat steps 4-10** for each subsequent marker in the panel.
12. **DAPI:** 1 ug/mL in PBS, 5 min, RT.
13. **Mount** with ProLong Gold antifade. Cure 24 h at RT in dark.

6.5 Imaging and Quantification

Imaging: Scan slides on Vectra Polaris (Akoya) or equivalent multispectral imaging system at 20x. Unmix spectra using single-stained controls.

Quantification (per mouse):

1. **DTC detection:** Count all pan-CK+ cells in 10 randomly selected 20x fields per section.
2. **Dormancy classification:** For each DTC, score:
 - NR2F1: positive (nuclear) or negative
 - Ki67: positive or negative
 - p27: positive (nuclear) or negative
 - **Dormant DTC = NR2F1+ AND Ki67- AND p27+**
 - **Proliferating DTC = Ki67+ AND (NR2F1- OR p27-)**
3. **Macrophage proximity:** Measure distance from each DTC to nearest CD68+ cell (or F4/80+ cell) using image analysis software (QuPath, HALO, or ImageJ).
4. **Minimum count:** Score at least 200 DTCs per condition for statistical power.

TIP

Use QuPath (free, open-source) for automated cell detection and classification. Train the classifier on manually annotated examples from 5-10 fields, then batch-process all sections.

Protocol 7: Liver Macrophage Flow Cytometry

7.1 Purpose

Quantify macrophage polarization (M1-like vs M2-like) and depletion efficiency in liver tissue.

7.2 Liver Non-Parenchymal Cell Isolation

1. After portal vein perfusion (Protocol 5.3), excise liver and mince with scissors into ~1 mm pieces in RPMI on ice.
2. Transfer to 15 mL tube with 5 mL digestion buffer:
 - RPMI + 0.05% collagenase IV + 0.01% DNase I + 2% FBS
3. Incubate 37°C, 30 min, shaking at 200 rpm.
4. Pass through 70 um cell strainer. Rinse with 5 mL RPMI + 2% FBS.
5. **Hepatocyte removal:** Centrifuge 50 x g for 3 min at 4°C. Discard pellet (hepatocytes). Collect supernatant (non-parenchymal cells).
6. **Percoll enrichment:** Resuspend cells in 3 mL 40% Percoll in PBS. Underlay with 3 mL 70% Percoll. Centrifuge 800 x g, 20 min, 4°C, no brake.
7. Collect interface layer (leukocyte-enriched). Wash with PBS + 2% FBS. Centrifuge 400 x g, 5 min.
8. Count cells. Expected yield: $2-5 \times 10^6$ leukocytes per liver.

7.3 Antibody Staining Panel

Marker	Fluorochrome	Clone	Purpose	Dilution
Live/Dead	Zombie NIR	-	Viability	1:1000
CD45	BV510	30-F11	Leukocytes	1:200
CD11b	BV605	M1/70	Myeloid cells	1:200
F4/80	APC-Cy7	BM8	Macrophages	1:100
Ly6G	PE-Cy7	1A8	Neutrophils (exclude)	1:200
Ly6C	FITC	HK1.4	Monocyte subsets	1:200
CD86	PE	GL-1	M1-like marker	1:100
CD206 (MMR)	APC	C068C2	M2-like marker	1:100
MHC-II	PerCP-Cy5.5	M5/114.15.2	Antigen presentation	1:200
CD64	BV410	X54-5/7.1	Macrophage vs monocyte	1:100
TIM4	BV711	RMT4-54	Kupffer cells	1:100
CD11c	BV785	N418	Dendritic cells / KCs	1:200

7.4 Staining Procedure

1. **Fc block:** Add anti-CD16/32 antibody (clone 93), 1:100 in PBS + 2% FBS, 10 min on ice.
2. **Live/Dead stain:** Add Zombie NIR, incubate 15 min on ice, protected from light.
3. **Wash** with PBS + 2% FBS.
4. **Surface antibody cocktail:** Add all surface antibodies in 100 uL staining buffer. Incubate 30 min on ice, protected from light.
5. **Wash** 2x with PBS + 2% FBS.
6. **Fix** with 1% PFA in PBS (if not proceeding to intracellular staining).
7. Acquire on flow cytometer (BD LSRFortessa, Cytex Aurora, or equivalent). Collect minimum 500,000 CD45+ events per sample.

7.5 Gating Strategy

1. **Singlets:** FSC-H vs FSC-A
2. **Live cells:** Zombie NIR-negative
3. **Leukocytes:** CD45+

4. **Myeloid cells:** CD11b+
5. **Exclude neutrophils:** Ly6G-negative
6. **Macrophages:** F4/80+ CD64+
7. **Kupffer cells vs recruited:** TIM4+ CD11c+ (Kupffer cells) vs TIM4- CD11c- (monocyte-derived)
8. **M1 vs M2:** Within macrophage gate, quantify:
 - M1-like: CD86+ MHC-II-high
 - M2-like: CD206+
 - **Report M1:M2 ratio** for each subset

TIP

Run Fluorescence Minus One (FMO) controls for CD86 and CD206 to set accurate gating boundaries. These markers have continuous expression, not bimodal.

Protocol 8: Single-Cell RNA Sequencing

8.1 Purpose

Profile transcriptomic states of DTCs and macrophages at single-cell resolution during each dormancy phase.

8.2 Sample Preparation

1. Isolate liver non-parenchymal cells (Protocol 7.2).
2. **FACS enrichment (required due to DTC rarity):**

Sort two populations per sample:

Population	Sort Gate	Target Count
DTCs	tdTomato+ / EpCAM+ / CD45- / Live	All recoverable (50-150)
Macrophages	CD45+ / CD11b+ / F4/80+ / Live	5,000-10,000

3. **Post-sort:** Collect sorted cells into RPMI + 20% FBS on ice. Count and assess viability (>85% required).
4. **Load on 10X Chromium:** Follow manufacturer's protocol for Chromium Next GEM Single Cell 3' v3.1.
 - DTC channel: Load all sorted DTCs
 - Macrophage channel: Load 10,000 cells
5. **Sequencing target:** 50,000 reads per cell, paired-end.

8.3 Key Timepoints

Day	Phase	Samples
7	Entry	3 mice per group
21	Established dormancy	3 mice per group
42	Maintenance	3 mice per group

8.4 Basic Analysis Pipeline

1. **Cell Ranger** (10x Genomics): Demultiplex, align to mouse genome (mm10) with GFP/tdTomato transgene reference added.
2. **Quality control (Seurat v5):**
 - Remove cells with <200 genes detected
 - Remove cells with >20% mitochondrial reads
 - Remove doublets (Scrublet or DoubletFinder)
3. **Integration:** Harmony or Seurat RPCA across samples/timepoints.
4. **Clustering:** Leiden algorithm, resolution 0.5-1.0.
5. **Cell type annotation:** SingleR + manual marker verification.
6. **Dormancy scoring:** Module score for Nr2f1, Bhlhe41, Cdkn1b (p27), Cdkn1a (p21).
7. **Macrophage polarization:** Module scores for M1 genes (Nos2, Tnf, Il12b, Cxcl9, Cd86) and M2 genes (Arg1, Mrc1, Chi3l3, Retnla, Ccl17).
8. **Ligand-receptor analysis:** NicheNet or CellPhoneDB v5.0 between macrophage and DTC populations.

WARNING

DTCs are extremely rare. If <30 DTCs are recovered per sample, pool 2-3 mice of the same group and timepoint (pseudobulk by group). Report the pooling strategy.

Protocol 9: Cytokine Profiling

9.1 Purpose

Measure M1-associated and M2-associated cytokines in liver tissue homogenates and serum.

9.2 Sample Collection

Serum:

1. Collect 500 uL blood by retro-orbital bleed or cardiac puncture at sacrifice.
2. Allow to clot 30 min at RT. Centrifuge 2000 x g, 10 min, 4°C.
3. Collect serum. Store at -80°C.

Liver homogenate:

1. Weigh 100 mg snap-frozen liver tissue.
2. Homogenize in 1 mL RIPA buffer + protease inhibitor cocktail + phosphatase inhibitors using bead homogenizer (TissueLyser, Qiagen).
3. Centrifuge 14,000 x g, 15 min, 4°C. Collect supernatant.
4. Measure total protein by BCA assay. Normalize all cytokine readings to mg total protein.

9.3 Multiplex Panel (Luminex or MSD)

Measure the following analytes:

Cytokine	M1/M2 Association	Relevance to Dormancy
IFN-gamma	M1	Dormancy entry (p38/STAT1 axis)
TNF-alpha	M1	Pro-inflammatory, NF-kB signaling
IL-12p70	M1	Th1 polarization
IL-10	M2	Immunosuppression
TGF-beta1	M2 (and others)	Dormancy exit (NRP2 axis)
TGF-beta2	M2 (and others)	Dormancy maintenance
IL-4	M2 polarizing	M2 induction
IL-13	M2 polarizing	M2 induction
CCL2/MCP-1	Monocyte recruitment	Macrophage recruitment
CXCL10/IP-10	M1	T cell chemoattractant
CSF-1	Macrophage survival	TAM maintenance
IL-6	Pro-inflammatory	Pleiotropic

9.4 Data Analysis

1. Generate standard curves per analyte using 5-parameter logistic (5PL) regression.
2. Report concentrations as pg/mg total protein (liver) or pg/mL (serum).
3. Compare across experimental groups by one-way ANOVA with Tukey post-hoc correction.
4. **Key comparison:** TGF-beta2 / TGF-beta1 ratio — should decrease with M2 impairment (Arm C).

Protocol 10: In Vitro Dormancy Co-Culture Validation

10.1 Purpose

Validate in vivo findings in a controlled system using Bone Marrow-Derived Macrophages (BMDMs) co-cultured with dormant MC38 cells.

10.2 BMDM Generation

1. Euthanize C57BL/6 mouse. Harvest femurs and tibiae.
2. Flush marrow with 10 mL cold PBS using 25G needle.
3. Pass through 70 μ m strainer. Centrifuge 300 x g, 5 min.
4. Resuspend in RBC lysis buffer, 2 min RT. Wash with PBS.
5. Culture at 1×10^6 cells/mL in DMEM + 10% FBS + 20 ng/mL M-CSF (recombinant mouse, PeproTech 315-02) on non-tissue-culture-treated Petri dishes.
6. Incubate 37°C, 5% CO₂ for 7 days. Add fresh M-CSF on Day 4.
7. Day 7: Verify >90% F4/80+ by flow cytometry. Harvest with cell scraper.

10.3 Macrophage Polarization

Phenotype	Stimulus	Concentration	Duration
M1 (classically activated)	LPS + IFN-gamma	100 ng/mL + 20 ng/mL	24 h
M2 (alternatively activated)	IL-4 + IL-13	20 ng/mL + 20 ng/mL	24 h
M0 (unpolarized)	M-CSF only	20 ng/mL	continued

10.4 Dormancy Induction in MC38

1. Seed MC38-Fucci2a cells at 5×10^4 per well in 24-well plate.
2. Induce dormancy: culture in DMEM + 0.5% FBS + 5 ng/mL TGF-beta1 for 72 h.
3. **Verify dormancy:** >80% Fucci-red (G1), Ki67-negative, p27-high by flow cytometry.

10.5 Co-Culture Setup

1. Plate 1×10^5 polarized BMDMs in lower chamber of 24-well transwell plate (0.4 μ m pore).
2. Plate 5×10^4 dormant MC38-Fucci2a cells in transwell insert (0.4 μ m pore allows soluble factor exchange, no direct contact).
3. **Co-culture conditions (n=4 per condition):**

Condition	Lower chamber	Insert	Purpose
M1 co-culture	M1 BMDMs	Dormant MC38	Does M1 maintain dormancy?
M2 co-culture	M2 BMDMs	Dormant MC38	Does M2 promote exit?
M0 co-culture	M0 BMDMs	Dormant MC38	Unpolarized control
No macrophages	Media only	Dormant MC38	Baseline

4. Readouts at 24, 48, 72 h:

- MC38 proliferation: EdU incorporation (Click-iT EdU Alexa Fluor 647, ThermoFisher)
- Fucci2a cell cycle: Flow cytometry for mCherry (G1) vs mVenus (S/G2/M)
- Dormancy markers: qRT-PCR for Nr2f1, Bhlhe41, Cdkn1b
- p-ERK/p-p38: Western blot

Protocol 11: Data Recording and Reporting

11.1 Required Records Per Mouse

Every mouse must have a record sheet containing:

Field	Details
Mouse ID	Unique identifier (e.g., A1-001)
Genotype	Wild-type, LysM-Cre x Nos2fl/fl, etc.
Birth date	dd/mm/yyyy
Injection date	dd/mm/yyyy
Cell dose	Number of cells injected
Treatment group	Clodronate, CpG-ODN, IL-4 complex, etc.
Treatment start date	dd/mm/yyyy
BLI data	Weekly total flux (photons/sec)
Health observations	Daily for first 7 days, then 3x weekly
Sacrifice date	dd/mm/yyyy
Tissue allocation	Which lobes to which analysis

11.2 Key Calculation Formulas

Dormancy Entry Rate (DER):

$$\text{DER} = (\text{Ki67- NR2F1+ GFP+ cells}) / (\text{total GFP+ cells}) \text{ at Day 7}$$

Dormancy Exit Rate (DXR):

$$\text{DXR} = (\text{DTCs exiting dormancy during observation}) / (\text{total dormant DTCs at start of observation})$$

M1:M2 Ratio:

$$\text{M1:M2} = (\text{CD86+ MHC-II-high F4/80+ cells}) / (\text{CD206+ F4/80+ cells})$$

p-ERK/p-p38 Ratio:

$$\text{Ratio} = (\text{mean fluorescence intensity of p-ERK}) / (\text{mean fluorescence intensity of p-p38}) \text{ per cell}$$

TIP

Threshold: $\text{p-ERK/p-p38} < 0.5$ = dormant. This threshold must be validated for MC38 cells using serum-starved (dormant) vs. log-phase (proliferating) controls.

11.3 Statistical Tests by Comparison

Comparison	Test	Correction
Dormancy fraction across groups	Fisher's exact test	Bonferroni (6 arms)
Time to metastasis	Kaplan-Meier + log-rank	Bonferroni
p-ERK/p-p38 ratio	Mann-Whitney U	Holm-Bonferroni
scRNA-seq DEGs	Wilcoxon rank-sum	Benjamini-Hochberg FDR
Cytokine levels	One-way ANOVA + Tukey	Tukey HSD
Longitudinal BLI	Mixed-effects model	FDR

Quick Reference: Experimental Arms Summary

Arm A — Total Macrophage Depletion

Group	Mice	Treatment	Start
A1	15	PBS liposomes (control)	Day 1
A2	15	Clodronate liposomes (early)	Day 1
A3	15	Clodronate liposomes (late)	Day 14
A4	15	Clodronate + CD169-DTR + DT	Day 14

Arm B — M1 Impairment (Dormancy Entry)

Group	Mice	Genotype/Treatment	Start
B1	15	LysM-Cre x Nos2fl/fl	Genetic
B2	15	LysM-Cre (Cre control)	Genetic
B3	15	Nos2fl/fl (no Cre control)	Genetic
B4	15	Anti-IFN-gamma Ab	Day -1
B5	15	Isotype control Ab	Day -1
B6	10	Anti-IFN-gamma Ab (late)	Day 7
B7	10	Isotype control Ab (late)	Day 7

Arm C — M2 Impairment (Dormancy Maintenance)

Group	Mice	Genotype/Treatment	Start
C1	15	LysM-Cre x Arg1fl/fl	Genetic
C2	15	LysM-Cre (Cre control)	Genetic
C3	15	Anti-IL-4Ralpha Ab	Day 14
C4	15	Isotype control Ab	Day 14
C5	15	CD169-DTR + DT	Day 14
C6	15	Cx3cr1-CreERT2 x DTA + tamoxifen	Day 14

Arm D — Temporal Switch (M2 -> M1)

Group	Mice	Treatment	Timing
D1	15	IL-4 complex -> IFN-gamma + LPS	Days 1-28 -> Days 28-42
D2	15	IL-4 complex (sustained)	Days 1-42
D3	15	IFN-gamma + poly(I:C) (sustained)	Days 1-42
D4	15	PBS vehicle	Days 1-42
D5	15	IL-4 complex -> IFN-gamma + anti-PD-1	Days 1-28 -> Days 28-42
D6	15	PI3K-gamma inhibitor (eganelisib)	Days 28-42

Appendix A: Master Acronym Glossary

Acronym	Full Term
AIW	Abdominal Imaging Window
ARG1	Arginase 1
BLI	Bioluminescence Imaging
BMDM	Bone Marrow-Derived Macrophage
BSL-2	Biosafety Level 2
CRC	Colorectal Cancer
CRLM	Colorectal Liver Metastasis
CSF1R	Colony Stimulating Factor 1 Receptor
CX3CR1	C-X3-C Motif Chemokine Receptor 1
DER	Dormancy Entry Rate
DMD	Dormancy Maintenance Duration
DT	Diphtheria Toxin
DTC	Disseminated Tumor Cell
DTR	Diphtheria Toxin Receptor
DXR	Dormancy Exit Rate
EGFP	Enhanced Green Fluorescent Protein
EpCAM	Epithelial Cell Adhesion Molecule
ERK	Extracellular Signal-Regulated Kinase
FACS	Fluorescence-Activated Cell Sorting
FBS	Fetal Bovine Serum
FDR	False Discovery Rate
Fucci2a	Fluorescent Ubiquitination-based Cell Cycle Indicator 2a
GFP	Green Fluorescent Protein
GJIC	Gap Junctional Intercellular Communication
HSC	Hepatic Stellate Cell
IF	Immunofluorescence
IFN-gamma	Interferon gamma
IL	Interleukin
iNOS	Inducible Nitric Oxide Synthase (NOS2)
IP	Intraperitoneal
IV	Intravenous
KC	Kupffer Cell
LPS	Lipopolysaccharide
LysM	Lysozyme M
M-CSF	Macrophage Colony-Stimulating Factor
MHC	Major Histocompatibility Complex
MMP	Matrix Metalloproteinase
MOI	Multiplicity of Infection
NOS2	Nitric Oxide Synthase 2 (iNOS)
NR2F1	Nuclear Receptor Subfamily 2 Group F Member 1
NRP2	Neuropilin 2
p-ERK	Phospho-Extracellular Signal-Regulated Kinase
p-p38	Phospho-p38
PD-1	Programmed Cell Death Protein 1
PDO	Patient-Derived Organoid
PI3K-gamma	Phosphoinositide 3-Kinase Gamma
RT	Room Temperature
SC	Subcutaneous
scRNA-seq	Single-Cell RNA Sequencing
STAT1	Signal Transducer and Activator of Transcription 1

Acronym	Full Term
TAM	Tumor-Associated Macrophage
TGF-beta	Transforming Growth Factor Beta
TLR	Toll-Like Receptor

Appendix B: Reagent and Vendor List

Reagent	Vendor	Catalog #	Notes
Clodronate liposomes	Liposoma BV	CP-005-005	Store 4°C, do NOT freeze
PBS liposomes	Liposoma BV	PB-005-005	Control
CpG-ODN 1826	InvivoGen	tlrl-1826	1 mg/mL in endotoxin-free H ₂ O
Recombinant mouse IL-4	BioLegend	574302	Aliquot, store -80°C
Anti-IL-4 Ab (clone 11B11)	BioLegend	504111	For IL-4 complex
Anti-IFN-gamma (clone XMG1.2)	Bio X Cell	BP0055	In vivo grade
Anti-IL-4Ralpha (clone M1)	Bio X Cell	-	In vivo grade
Isotype control (rat IgG1)	Bio X Cell	BP0088	Control for XMG1.2
D-Luciferin potassium salt	GoldBio	LUCK-1G	15 mg/mL in PBS
Diphtheria toxin	Sigma	D0564	EXTREMELY toxic
Pexidartinib (PLX3397)	Selleckchem	S7818	PI3K-gamma inhibitor
Collagenase IV	Worthington	LS004188	Liver digestion
DNase I	Roche	10104159001	Liver digestion
Percoll	Cytiva	17-0891-01	40%/70% gradient
Zombie NIR viability dye	BioLegend	423105	Live/Dead for flow
Anti-CD16/32 (clone 93)	BioLegend	101319	Fc block
Opal 7-plex kit	Akoya	NEL811001KT	Multiplex IF
Buprenorphine	Patterson Veterinary	07-890-6657	Analgesia
Isoflurane	VetOne	501017	Anesthesia

Appendix C: Troubleshooting Guide

Problem	Possible Cause	Solution
Low DTC yield at Day 7	Injection technique (backflow), cell viability	Practice slow injection (30 sec); verify >95% viability; try portal vein injection instead
No BLI signal	Luciferase reporter loss, insufficient cells	Re-check GFP expression before injection; increase cell dose in pilot
Clodronate depletion <70%	IV injection failure (tail vein misses)	Switch to IP route, 3x weekly; verify technique with trypan blue test injection
High background in IF	Incomplete stripping between Opal rounds	Extend microwave time to 25 min; verify stripping with secondary-only control
Low macrophage yield from liver	Incomplete digestion, hepatocyte contamination	Increase collagenase IV to 0.1%; ensure hepatocyte spin (50 x g) removes parenchymal cells
Fucci2a signal lost	Transgene silencing	Re-sort for Fucci-positive cells; use early passage cells (<P20 post-engineering)
DT toxicity in CD169-DTR mice	Overdosing	Confirm dose calculation; never exceed 10 ng/g; use wild-type + DT controls
Poor scRNA-seq viability	Processing delay, mechanical stress	Process within 30 min; use gentle pipetting; keep cold; add 0.04% BSA to PBS
Variable BLI between mice	Inconsistent luciferin timing	Standardize: always image at exactly 10 min post-luciferin injection; same person injects and times
M1:M2 ratio doesn't shift with CpG/IL-4	Dose insufficient, mouse strain background	Titrate CpG up to 100 ug; verify IL-4 complex formation (37°C, 30 min pre-incubation)